

Models and Tools for Cyber-Physical Systems

Chapter 9: Hybrid systems

Instructors: Suman Sourav, Jonathan Julián Huerta y Munive
{sumansourav,jjhymuni}@cs.aau.dk

Original slides courtesy of Rajeev Alur
alur@cis.upenn.edu

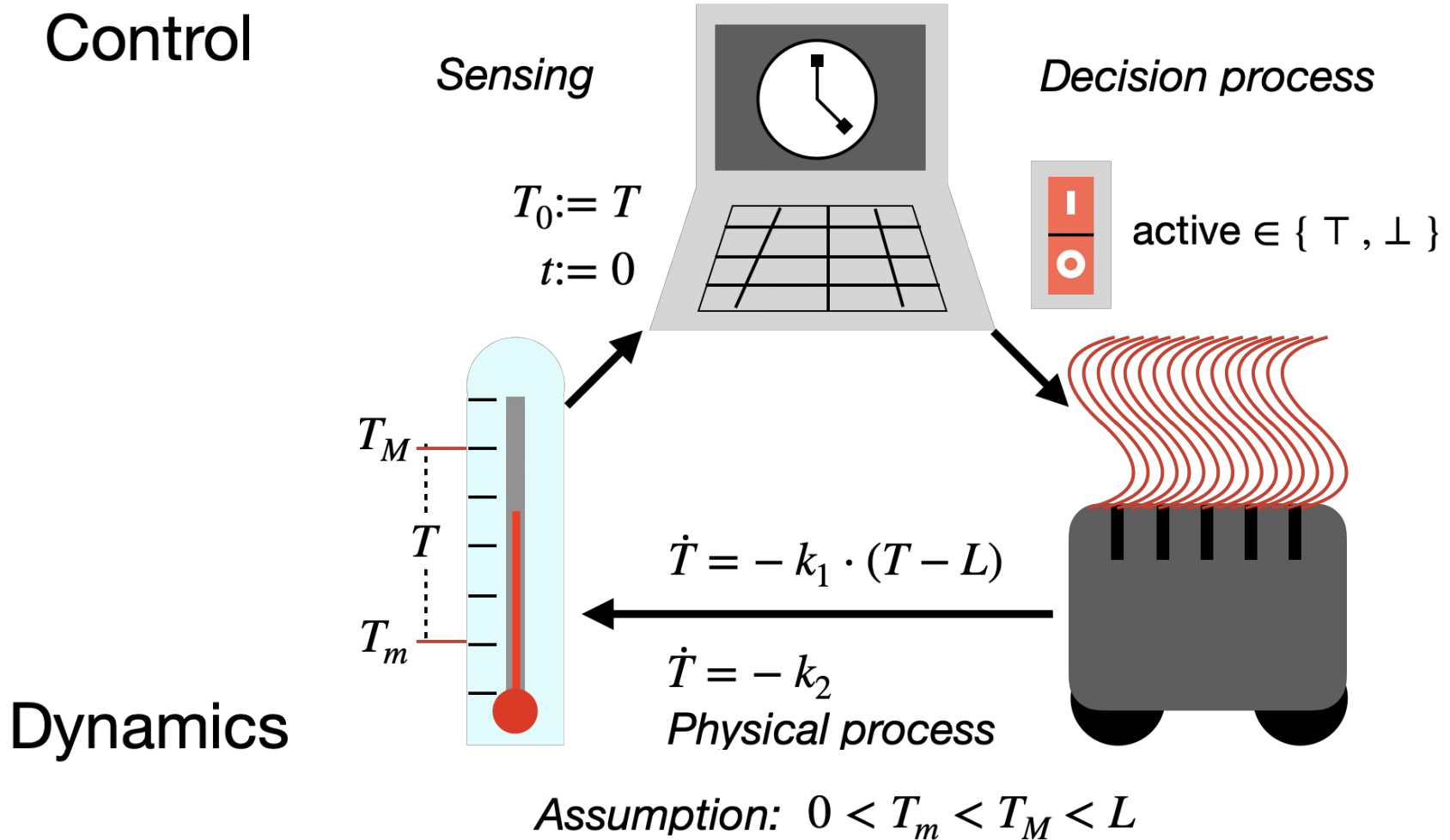
Today's schedule

- ☐ Example of hybrid system: thermostat
- ☐ Definition of hybrid process
- ☐ Exercise to practice designing a hybrid process
- ☐ Exercise to understand Zeno behaviour
- ☐ Observation on the stability of hybrid systems
- ☐ Optionals:
 - ☐ Research example
 - ☐ Challenge exercise: reactor

Models of Reactive Computation

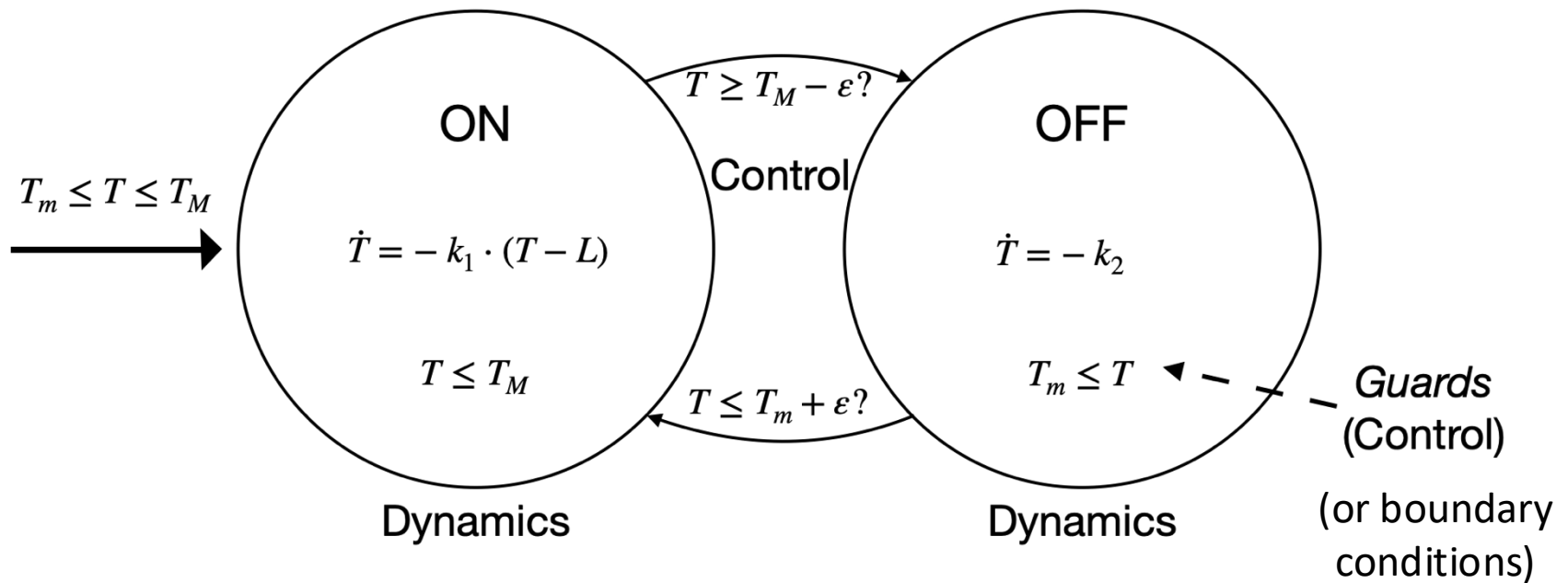
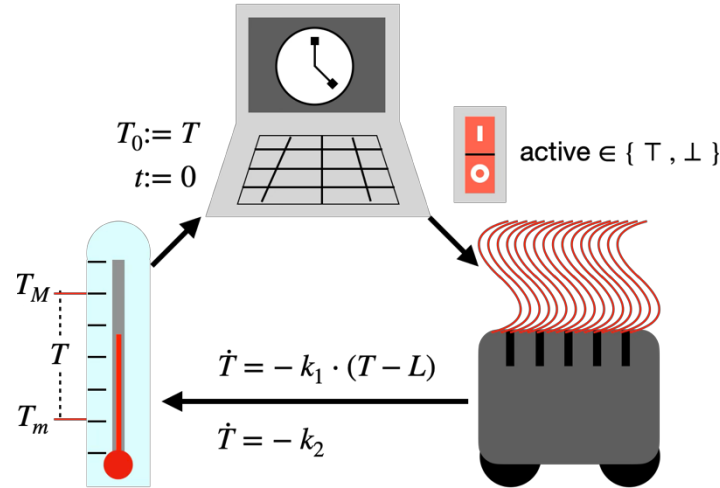
- ❑ Continuous-time model for dynamical system
 - Synchronous, where time evolves continuously
 - Execution of system: Solution to algebraic / differential equations
- ❑ Timed model
 - Like asynchronous for communication of information
 - Clocks evolve continuously, and constraints on delays allow synchronous/global coordination
- ❑ Hybrid systems
 - Generalization of timed processes
 - During timed transitions, evolution of state/output variables specified using differential equations as in dynamical systems

Example: Thermostat Hybrid System

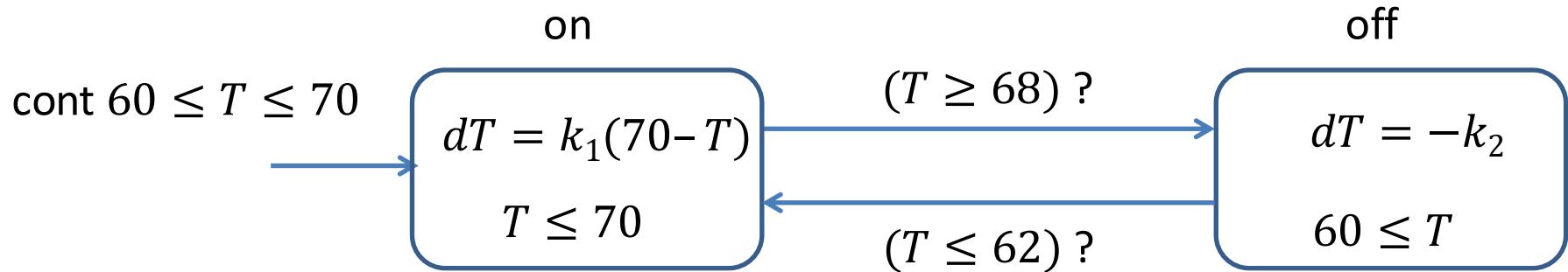


Objective: model the cyber-physical system with a mathematical abstraction

Example: Thermostat Hybrid System



Self-Regulating Switching Thermostat



State machine with two modes +

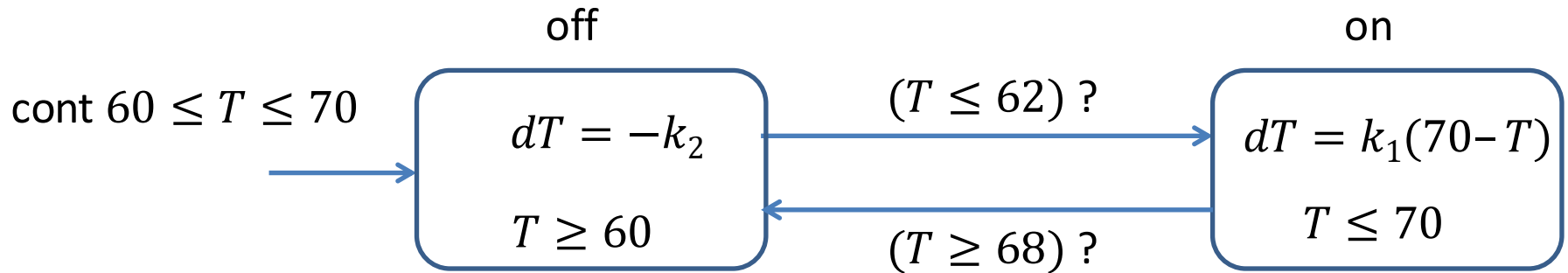
State variable T to model temperature: type **cont**

T can be tested and updated during mode-switches

T changes continuously during timed transitions given by differential equations

Invariants (as in timed model) constrain how long can a timed transition be

Executions of Thermostat



Initial state = (off, T_0) with T_0 in the interval $[60, 70]$

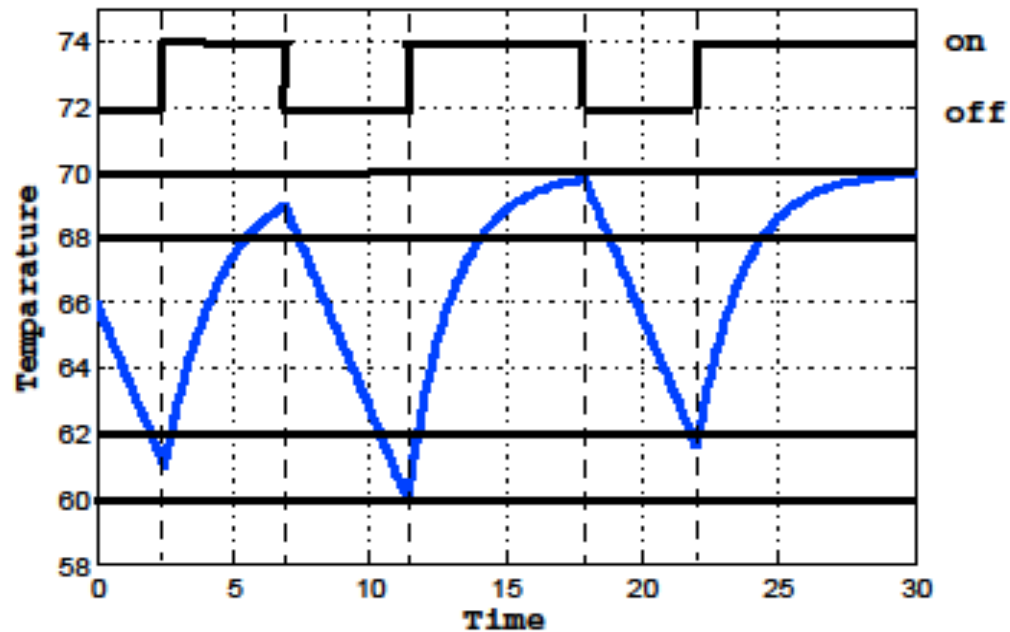
During a timed transition, T decreases continuously: $T(t) = T_0 - k_2 t$

Mode-switch to on enabled when $T \leq 62$, and must happen before T reaches 60

As time elapses in mode on, T increases according to $T(t) = 70 - (70 - T^*)e^{-k_1(t-t^*)}$,
 t^*, T^* : time and temperature upon entry to mode on

Mode-switch to off enabled when $T \geq 68$, and must happen before T reaches 70

Simulation Plot of an Execution



Python Instructions for those new to it

- Like for many other languages, it is convenient to have a package manager:
 - pip -> official Python package manager
 - (mini)conda -> Anaconda, Inc's package manager for data science and AI (slow and a bit behind pip's latest additions).
 - (micro)mamba -> Efficient reimplementation of conda (uses SAT-solving for dependency resolution)
- It is good practice to create "environments" with specific packages for specific projects (no global installs):
 - "micromamba create -n env_name python=N -c channel_name"
 - "micromamba activate env_name"
 - "conda create -n env_name python=N -c channel_name"
 - "conda activate env_name"
- For this course, install standard data-science and plotting libraries:
 - "micromamba install matplotlib"
 - "pip install matplotlib"
- Run the scripts
 - "python /path/to/my_script.py"

Definition of a Hybrid Process

Definition of Hybrid Process: Syntax

- A hybrid process HP consists of
 1. An asynchronous process P , where some of the state variables can be type cont (ranging over real numbers)
 2. A continuous-time invariant CI which is a Boolean expression over the state variables of P
 3. For every output variable y of type cont , a Lipschitz-continuous real-valued expression that gives the value of y as a function of state variables and continuous input variables
 4. For every state variable x of type cont , a Lipschitz-continuous real-valued expression that gives the rate of change of x as a function of state variables and continuous input variables

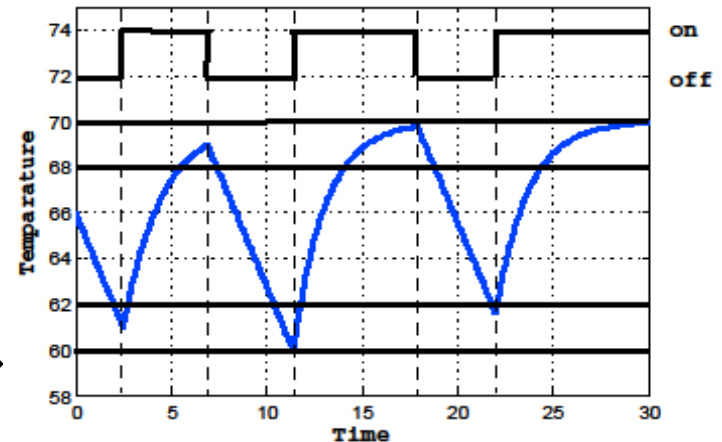
Definition of Hybrid Process: Semantics

- ❑ Define inputs, outputs, states, initial states, internal actions, input actions, output actions exactly the same as the asynchronous model
- ❑ Timed actions: Given a state s and real-valued time $\delta > 0$ and a continuous input signal $u(t)$ that gives values for continuous inputs over time interval $[0, \delta]$, the corresponding state/output signal over $[0, \delta]$ is uniquely defined so that
 1. Initial state $s(0)$ equals s
 2. Discrete (i.e. non-cont) state variables stay unchanged
 3. For each continuous output variable y , the value $y(t)$ satisfies the corresponding algebraic equation
 4. For each continuous state variable x , the derivative $dx(t)/dt$ of the signal satisfies the corresponding differential equation
 5. At all times t in $[0, \delta]$, the signal value $s(t)$ satisfies the invariant constraint CI

Executions of Hybrid Processes

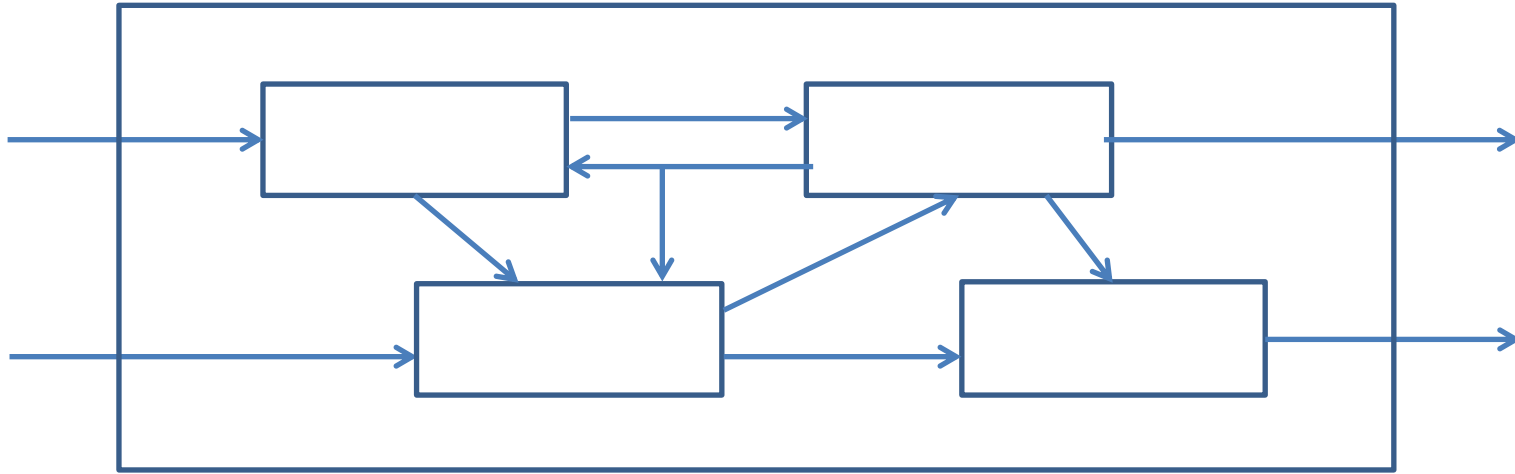
Starting from an initial state, execute either a discrete step (input, or output or internal action) or a timed step (need to solve system of differential equations)

$$\begin{aligned} &(\text{off}, 66) \xrightarrow{2.5} (\text{off}, 61) \xrightarrow{\varepsilon} (\text{on}, 61) \xrightarrow{3.7} \\ &(\text{on}, 69.02) \xrightarrow{\varepsilon} (\text{off}, 69.02) \xrightarrow{4.4} (\text{off}, 60.22) \xrightarrow{\varepsilon} \\ &(\text{on}, 60.22) \xrightarrow{7.6} (\text{on}, 69.9) \xrightarrow{\varepsilon} (\text{off}, 69.9) \xrightarrow{4.1} \\ &(\text{off}, 61.7) \xrightarrow{\varepsilon} (\text{on}, 61.7) \dots \end{aligned}$$



Concepts based on transition systems such as reachable states, safety and liveness requirements, all apply to hybrid systems

Block Diagrams



- ❑ Component processes can now be hybrid processes
 - Need to define Instantiation, Composition, Output Hiding
- ❑ Channels connecting processes of two types
 1. Sender/receiver communication of values during discrete steps as in the asynchronous model
 2. Continuously evolving signals during timed steps as in the model of continuous-time dynamical systems

Summary of the Model

- ❑ Generalizes timed model
 - Variables evolving continuously during a timed action can have complex dynamics, clocks being a very special case
- ❑ Generalizes continuous-time dynamical systems
 - Discontinuous changes to system state now can be modeled
- ❑ Generalizes asynchronous model
 - Distributed/multi-agent systems can be modeled
- ❑ Suitable for modeling of cyber-physical systems (in full generality)
- ❑ Existing commercial tool support: Modelica, Stateflow/Simulink
- ❑ Challenge for analysis
 - Even if dynamics in individual modes is linear, due to discrete changes, not possible to obtain closed-form solutions, or general theorems about stability

Practice exercise: write a hybrid automata for a water tank

